FROM THE LANDSCAPE TO THE PLANT: INTEGRATING, VISUALIZING AND ANALYZING PLANT ENVIRONMENT, PHENOTYPE AND GENOTYPE USING CARTOGRAPLANT, WILDTYPE AND TRIPAL GALAXY

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DATA INTEGRATION

RAW DATA + METADATA+ ONTOLOGIES + STANDARDS
GENOTYPE (G) x PHENOTYPE (P) x ENVIRONMENT (E)

1. RAW DATA → GENOTYPE/ PHENOTYPE

Biocuration efforts of affiliated databases

TreeGenes, BIEN, DRYAD, TreeSnap

Direct submission of studies: Tripal Plant PopGen Submit (TPPS) Pipeline

Drupal module which **captures data and metadata describing genotype, phenotype and environmental studies**, using a series of questions to properly describe the experimental design.

WildType (available soon)

ENVIRONMENTAL LAYERS

- CLIMATE DATA.
- ECOREGIONS
- SOIL TYPE
- FOREST FRAGMENTATION
- NEON FIELD STATIONS
- SEED ZONES
- BIODIVERSITY HOTSPOTS
- AGRICULTURE AREAS
- SPECIES RANGES
- PET/ARIDITY
- LAND COVER
- FOREST CANOPY
- BIOTIC DAMAGE
- NDVI (PLANT HEALTH)
- HUMAN IMPACT
- PROTECTED AREAS

2. METADATA + ONTOLOGIES + STANDARDS → miappe

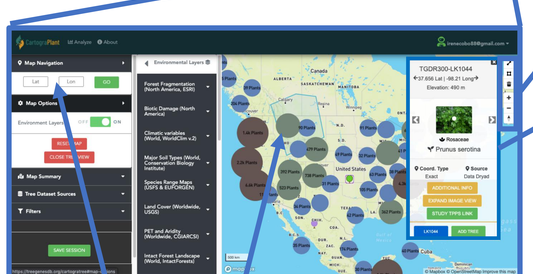
Ensures the **FAIR**ness of data:
Findable, Accessible, Interoperable, Reusable

Metadata collected using standards and associated ontologies following the Minimal Information About a Plant Phenotyping Experiment (MIAPPE)

DATA VISUALIZATION

LEFT PANEL
Action panel to interact with the map and plants, located to the left of the screen

RIGHT PANEL
An interactive map, showing the selected plants and environmental layers on the left panel



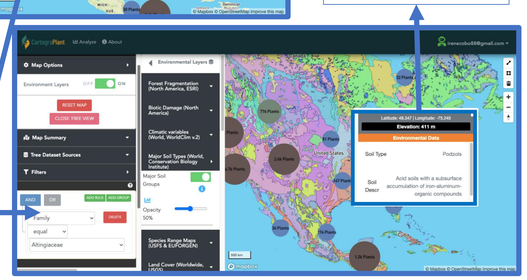
Zoom/polygon drawing selection

By clicking on a plant, detailed information is displayed

By clicking on the map, information about the environmental layer is displayed

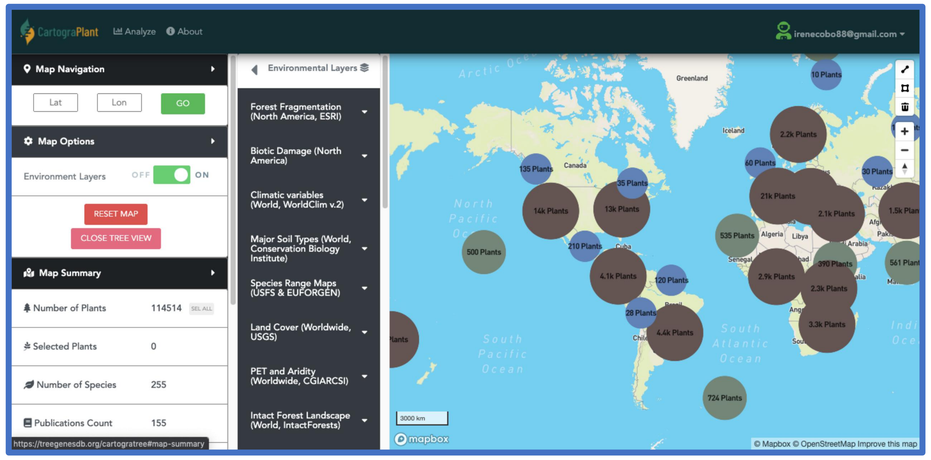
Users can **navigate** on the map by using the **zoom buttons** or by **typing the coordinates of interest**

Plants can be selected for data analysis **by** multiple **filtering** criteria (taxonomy, study, data source…) or **by drawing the area of interest directly on the map**



INTRODUCTION

The **Tripal platform** (https://tripal.info/) **provides a mechanism by which online genomic, genetic and breeding databases can reduce duplication of effort, develop using a common platform, and improve the sustainability of respective online databases**. To this end, Tripal stores data **following the FAIR principles: Findability, Accessibility, Interoperability and Reusability**. It is developed using the Drupal Content Management System (CMS) and the Chado relational database schema. The software is community maintained and supports 29 databases around the world housing genetic, genomic, and breeding for thousands of species through databases such as TreeGenes (Falk et al. 2018, https://treegenesdb.org/), Hardwood Genomics Web (HGW, http://www.hardwoodgenomics.org/), and Genome Database for Rosaceae (GDR, http://www.rosaceae.org). **The core functionalities of Tripal are extended through extension modules.**
One such module is CartograPlant (https://treegenesdb.org/ct), **a web application that integrates, visualizes, and analyzes genotypic, phenotypic, environmental data from georeferenced plants.**
TreeSnap (https://treesnap.org/) is a **mobile application** that **engages informed, taxonomically trained volunteers to identify and submit usable data for tree research programs**, such as GPS locations, images, and observations (traits) of interest to scientists. TreeSnap is in the process of providing a **more customized platform** where **scientists** can define the constraints of data collection, **WildType,** which is targeted towards landscape-based studies (as opposed to row-crop style investigations). **The phenotypic data collected through TreeSnap (for citizen scientists) and WildType (for researchers) can be also available in CartograPlant for data analysis.**



Currently, CartograPlant hosts 155 studies and integrates genotypic and phenotypic data from 8.3 M georeferenced plants with 710 environmental layers.

CartograPlant also allows data analysis by providing appropriate HPC-backed workflows through Galaxy thanks to the Tripal Galaxy module

Why is this relevant and timely?

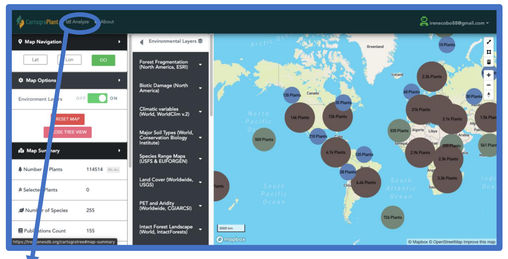
- Climate change is threatening plant health and productivity
- It is unknown whether plant breeding can keep pace with the rate and direction of environmental change
- The frequency and impact of invasive pests and pathogens is increasing due to globalization and climate change

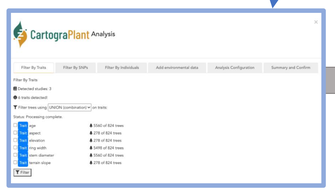
Conclusion

Future plant health and productivity will depend on the match between genotypes (and their resulting phenotypes) and new environments, necessary to **identify genes controlling traits that provide tolerance to biotic and abiotic stresses.** Thus, **tools that collect, integrate, and facilitate this assessment** (e. g. data analysis) **are critical.**

DATA ANALYSIS

1. After the data **selection**, users can perform data analysis by clicking "Analysis". Prior to data analysis, the **data** can be **filtered by trait, SNP and individual**. **Environmental data and the analytic workflow** can be also **selected** at this stage.



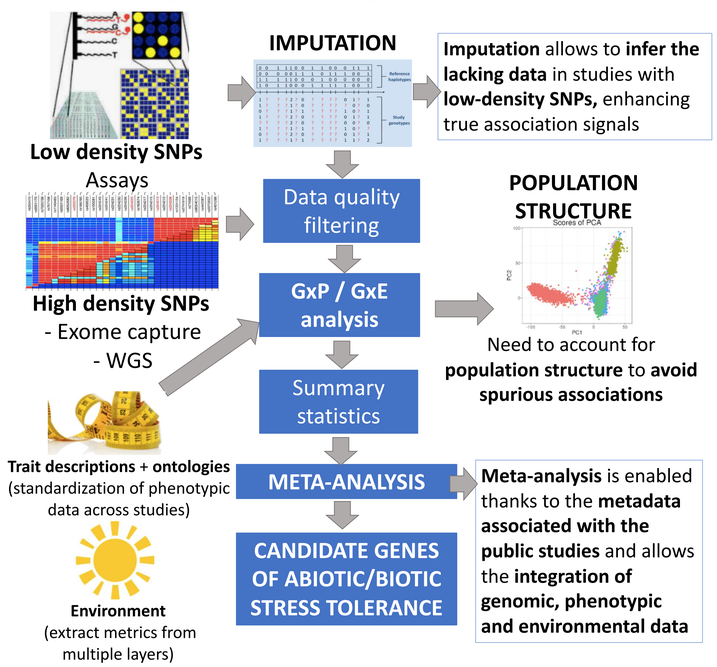


2. Once the data is **filtered**, users can run the selected **analytic workflows on the filtered data** from CartograPlant **in Galaxy**, thanks to the **Tripal Galaxy module.**

FLEXIBLE ANALYTIC WORKFLOWS
G x P (association genetics)
G x E (landscape genomics)
Population structure
Meta-analysis

ANALYTIC WORKFLOW: IMPUTATION-BASED META-ANALYSIS FOR GXP AND GXE ANALYSIS

This workflow **takes advantage of the strengths of CartograPlant**: the **diversity of data types** (genotypic, phenotypic and environmental), **metadata collection** and robust **integration of regional and global environmental layers**. ***Populus trichocarpa*** is used as a **model** (compact genome size, economic and ecological importance, and high number of available genomic and phenotypic resources).



TreeGenes · Tripal · GMOD · AgBioData · DBTREES · GTTN · Galaxy · USDA · CartograPlant